



CANDIDATE
NAME

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9709/33

May/June 2026

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

[illegible]



- 2 Find the set of values of x satisfying the inequality $|3^{2x-1} - 4| > 2$. Give your answers correct to 3 significant figures.

[4]

[illegible]

- 3 On an Argand diagram, shade the region which represents the complex numbers, z , satisfying the inequalities $|z - 2 + i| \leq 3$ and $|z| \geq |z - 2 + 2i|$. [5]



[6]

[illegible]



5 (a) The parametric equations of a curve are

$$x = \tan 2t, \quad y = \frac{1}{\sin 2t}.$$

Show that $\frac{dy}{dx}$ can be written as $\frac{A \cos 2t}{\tan^2 2t}$, where A is a constant to be found. [5]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are faint, diagonal watermarks visible across the page that read "PapaCambridge.com".



(b) Find an exact equation of the tangent to the curve at the point where $t = \frac{1}{8}\pi$.

[4]

[illegible]



6 Use the substitution $u = \sqrt{x}$ to find the exact value of $\int_{\frac{1}{16}\pi^2}^{\frac{1}{4}\pi^2} \frac{\cos(3\sqrt{x})}{\sqrt{x}} dx$. [6]

[illegible]



- 7 The square roots of $3 - (20\sqrt{7})i$ can be expressed in the Cartesian form $x + iy$, where x and y are real and exact.

By first forming a quartic equation, find the square roots of $3 - (20\sqrt{7})i$. Give your answer in exact Cartesian form. [5]

[5]

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- 8 (a) By sketching a suitable pair of graphs, show that the equation $x^3 = 4 - 2x$ has exactly one real root.

[1]

- (b) Show that, if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{3x_n^3 + 4}{4x_n^2 + 2}$$

converges, then it converges to the root of the equation in 8(a).

[2]

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(c) Show by calculation that this root lies between 1 and 2.

[2]

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(d) Use the iterative formula given in **8(b)** to calculate this root correct to 2 decimal places. Give the result of each iteration to 5 decimal places.

[3]

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9 Let $f(x) = \frac{36 - 4x - 3x^2}{(3x + 1)(x^2 + 4)}$.

(a) Express $f(x)$ in partial fractions.

[5]

[illegible]



(b) Hence, obtain the expansion of $f(x)$ in ascending powers of x , up to and including the term in x^3 . [4]

(c) State the set of values of x for which the expansion is valid. [1]



10 The variables x and y satisfy the differential equation

$$4e^{2x} \frac{dy}{dx} = x(y+3)$$

for $y > -3$. It is given that $y = -2$ when $x = 0$.

Solve the differential equation and obtain an expression for y in terms of x .

[8]

This image shows a full page of blank handwriting practice paper. It features multiple sets of horizontal lines. Each set consists of a solid top line, a dashed midline, and a solid bottom line, providing a guide for letter height and placement. The paper is white and contains no other markings or text.

DO NOT WRITE IN THIS MARGIN



Handwriting practice area with 20 horizontal dotted lines.





- 11 The points A and B have position vectors $\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ -3 \\ 5 \end{pmatrix}$ respectively. The line t passes through the points A and B .

(a) Find a vector equation for t .

[2]

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The line s has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$.

(b) Find the acute angle between s and t .

[3]

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17

(c) Find the length of the perpendicular from A to s .

[5]





If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]

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